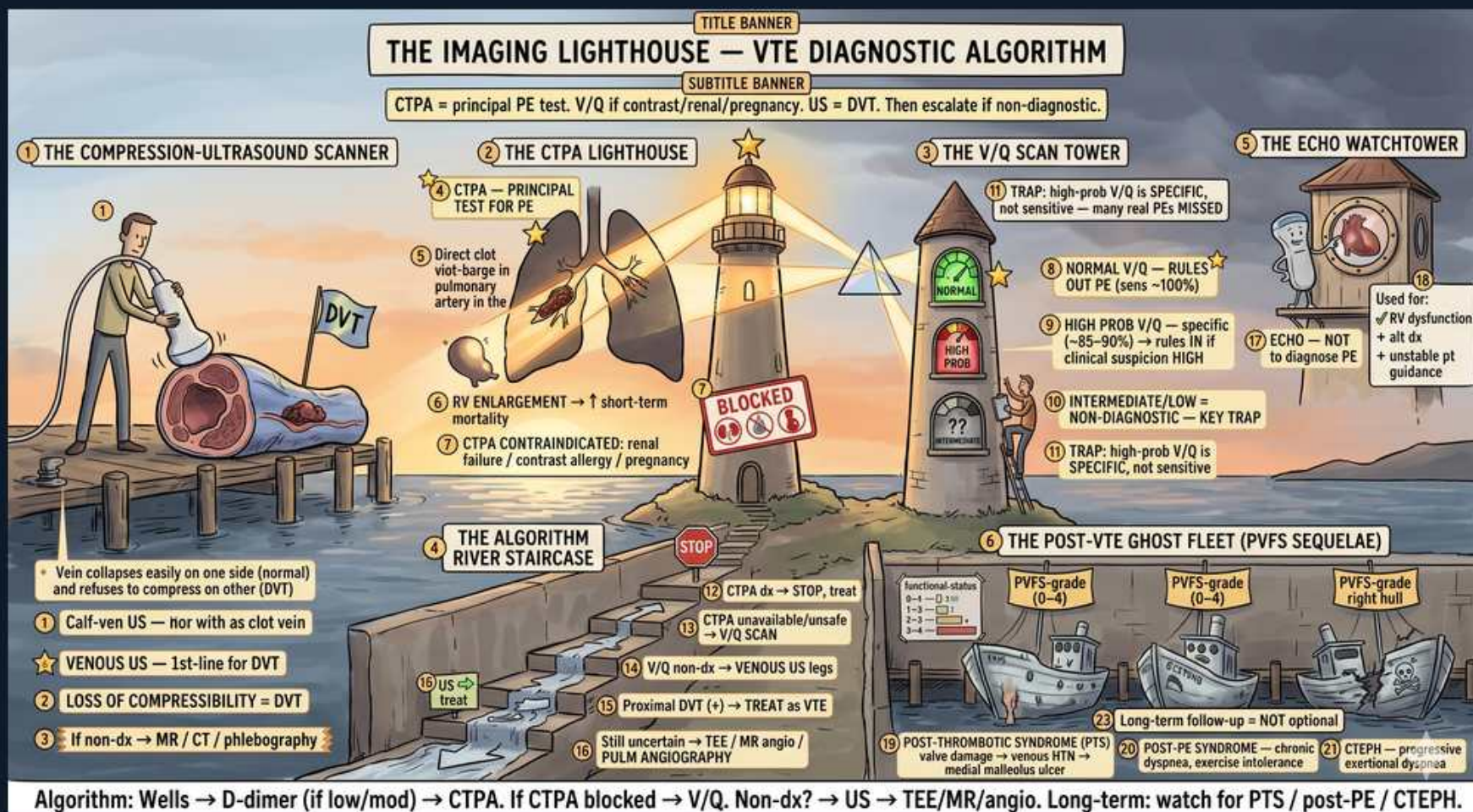


VTE — Diagnostic Imaging Algorithm (CTPA, V/Q, US, Echo)



1. Compression ultrasound

WHAT YOU SEE

Compression-ultrasound scanner on a leg vein

WHAT IT ENCODES

Compression ultrasound with Doppler is the first-line, non-invasive test for suspected DVT, examining the common femoral, femoral, and popliteal veins. The single diagnostic criterion is loss of vein compressibility under probe pressure (a normal vein collapses, a clot-filled vein does not). Sensitivity/specificity for proximal DVT are >95%, but it is less reliable for isolated calf vein DVT, where serial scans may be needed. Mnemonic: 'If it won't squish, there's a clot in the dish.'

BOARD TRAP

WRONG: 'Order a D-dimer to confirm a DVT already seen as a non-compressible femoral vein on US.' D-dimer is a rule-OUT test for low pretest probability, not a confirmatory test — a positive compression US already establishes DVT and warrants anticoagulation.

2. Loss of compressibility = DVT

WHAT YOU SEE

Vein that won't collapse under probe

WHAT IT ENCODES

A vein that fails to fully collapse when the transducer compresses it is the defining sign of an intraluminal thrombus; color Doppler showing absent or filling-defect flow is supportive. This is operator-dependent and limited in the pelvic/iliac veins and in obese or edematous legs, where CT/MR venography may be required. Mnemonic: 'Compressible = clear; non-compressible = clot.'

BOARD TRAP

WRONG: 'A normal compression US of the leg rules out PE.' A negative leg US does not exclude PE — the clot may have already embolized or arisen elsewhere; if PE is suspected with negative legs, proceed to CTPA/V/Q based on pretest probability and D-dimer.

3. CTPA = principal PE test

WHAT YOU SEE

Lighthouse labeled CTPA principal imaging test

WHAT IT ENCODES

CT pulmonary angiography is the first-line, gold-standard imaging test for PE, directly visualizing filling defects in the pulmonary arteries down to the segmental/subsegmental level. It is fast, widely available, and simultaneously evaluates alternative diagnoses (pneumonia, dissection). It requires IV iodinated contrast and radiation. Mnemonic: 'CTPA = the lighthouse you steer toward for PE.'

BOARD TRAP

WRONG: 'Send a hemodynamically unstable, hypotensive patient with suspected massive PE to CTPA first.' An unstable patient may not survive transport to CT — bedside echocardiography showing RV strain (plus clinical context) supports empiric thrombolysis; CTPA is for stable patients.

4. RV enlargement on CTPA

WHAT YOU SEE

Image note: RV enlargement → short-term mortality

WHAT IT ENCODES

On CTPA, an RV:LV diameter ratio >1.0 (or >0.9) on the axial four-chamber reconstruction signals RV pressure overload and predicts elevated 30-day/short-term mortality. Combined with elevated troponin/BNP, it reclassifies a normotensive PE as intermediate-HIGH risk, prompting closer monitoring and consideration of reperfusion if the patient deteriorates. Mnemonic: 'Big RV on CT = bad prognosis flag.'

BOARD TRAP

WRONG: 'RV enlargement on CTPA in a normotensive patient means give thrombolytics now.' Imaging RV strain alone defines intermediate-risk PE — systemic thrombolysis is reserved for hemodynamic instability (or decompensation); intermediate-risk patients are usually anticoagulated and monitored (PEITHO showed excess major bleeding/stroke with routine lysis).

5. CTPA contraindicated

WHAT YOU SEE

Blocked road sign: renal failure / contrast allergy / pregnancy

WHAT IT ENCODES

CTPA should be avoided when IV iodinated contrast is hazardous: significant renal impairment (contrast nephropathy), severe contrast allergy/anaphylaxis history, and relative avoidance in pregnancy/young women to limit breast radiation. In these cases the algorithm escalates to a V/Q scan, which uses no iodinated contrast. Mnemonic: 'Kidney, allergy, pregnancy → road blocked → reroute to V/Q.'

BOARD TRAP

WRONG: 'A creatinine of 3.5 means you cannot image for PE at all.' Renal failure contraindicates contrast CTPA but not diagnosis — a V/Q scan (no iodinated contrast) is the correct alternative; missing a PE is far more dangerous than the limitation.

6. V/Q scan tower

WHAT YOU SEE

Tower labeled V/Q scan, second-line

WHAT IT ENCODES

The ventilation–perfusion scan is the principal alternative to CTPA when contrast is contraindicated (renal failure, contrast allergy) or to reduce radiation in young/pregnant patients. It compares inhaled and IV radiotracer distributions; a PE shows perfusion defects in normally ventilated lung (mismatch). It performs best with a normal baseline chest X-ray. Mnemonic: 'V/Q = the backup tower when CTPA road is blocked.'

BOARD TRAP

WRONG: 'V/Q is useless in a patient with COPD/abnormal CXR, so just give contrast despite the renal failure.' V/Q is less interpretable with parenchymal disease but is still the contrast-free option; alternatives include lower-extremity US or careful risk-benefit CTPA — not forcing contrast in renal failure by default.

7. Normal V/Q rules out PE

WHAT YOU SEE

Green light: normal V/Q ~100% rules out PE

WHAT IT ENCODES

A completely NORMAL perfusion scan has a very high negative predictive value (~100%) and effectively excludes clinically significant PE, allowing anticoagulation to be safely withheld. This mirrors how a negative high-sensitivity D-dimer rules out PE in a low/intermediate pretest patient. Mnemonic: 'Normal V/Q = green light, PE excluded.'

BOARD TRAP

WRONG: 'A low-probability (not normal) V/Q scan rules out PE.' Only a NORMAL perfusion scan excludes PE — 'low probability' still carries residual risk and must be combined with pretest probability/D-dimer; do not equate low-probability with normal.

8. High-prob V/Q = specific

WHAT YOU SEE

Note: high-probability V/Q specific (~85-90%) but not sensitive

WHAT IT ENCODES

A high-probability V/Q scan (≥ 2 large mismatched segmental perfusion defects) is highly specific (~85–90%) and, with a concordant high pretest probability, rules PE IN and justifies treatment. However V/Q is not very sensitive overall and produces many non-diagnostic results, so a non-high-probability scan cannot exclude PE. Mnemonic: 'High-prob V/Q rules IN; it does not rule out.'

BOARD TRAP

WRONG: 'A non-high-probability V/Q scan in a high pretest-probability patient excludes PE.' The discriminator is pretest probability — with high clinical suspicion, a non-high-prob scan is non-diagnostic and you must proceed to CTPA, US, or angiography rather than stopping.

9. Intermediate/non-diagnostic V/Q trap

WHAT YOU SEE

Yellow light: intermediate/non-diagnostic V/Q

WHAT IT ENCODES

Intermediate/indeterminate V/Q results are common (especially with abnormal CXR or COPD) and neither confirm nor exclude PE. The correct move is further testing: bilateral leg compression US (a positive proximal DVT confirms VTE and ends the workup) or CTPA/pulmonary angiography. Mnemonic: 'Yellow light = keep going, don't stop the workup.'

BOARD TRAP

WRONG: 'An intermediate/non-diagnostic V/Q in a symptomatic patient lets you safely stop and discharge.' Indeterminate scans require escalation (leg US, CTPA, or angiography) — stopping risks missing a treatable, potentially fatal PE.

10. Echo watchtower

WHAT YOU SEE

Watchtower labeled echocardiography

WHAT IT ENCODES

Echocardiography does NOT diagnose PE — it risk-stratifies. In an unstable patient it can rapidly show signs of acute RV pressure overload (RV dilation, septal flattening/D-sign, McConnell's sign of apical sparing with free-wall hypokinesis, TR with elevated PASP), supporting empiric thrombolysis when CTPA is unsafe. Mnemonic: 'Echo = watchtower for RV strain in the crashing patient.'

BOARD TRAP

WRONG: 'A normal echocardiogram rules out PE.' Echo is neither sensitive nor specific enough to exclude PE (many PEs have normal echoes) — its role is to detect RV strain for risk stratification in unstable patients, not to confirm or exclude the diagnosis.

11. Wells → D-dimer (algorithm)

WHAT YOU SEE

River staircase step: Wells then D-dimer if low/moderate

WHAT IT ENCODES

Start with the Wells (or revised Geneva/PERC) score to set pretest probability. For LOW/moderate probability, a high-sensitivity D-dimer below the (age-adjusted: $\text{age} \times 10$ if >50) cutoff has high negative predictive value and rules OUT PE with no imaging. A positive D-dimer, or HIGH pretest probability, goes straight to CTPA. Mnemonic: 'Low Wells + negative D-dimer = done; high Wells = skip D-dimer, image.'

BOARD TRAP

WRONG: 'Order a D-dimer in a patient with high pretest probability (Wells >6) for suspected PE.' D-dimer is a rule-out tool only for low/intermediate probability — in high pretest probability its negative result is unreliable, and a positive result is expected and unhelpful; proceed directly to CTPA.

12. CTPA → if blocked → V/Q

WHAT YOU SEE

Staircase: CTPA, if unavailable/contraindicated → V/Q

WHAT IT ENCODES

When imaging is indicated, CTPA is the default; if it is contraindicated (renal failure, contrast allergy, pregnancy) or unavailable, escalate to a V/Q scan, and if that is non-diagnostic, use leg compression US or formal pulmonary angiography. D-dimer is unhelpful once the decision to image has been made. Mnemonic: 'CTPA first; blocked road → V/Q → US/angio.'

BOARD TRAP

WRONG: 'If CTPA is contraindicated, give the contrast anyway because there is no other PE test.' V/Q (and leg US/angiography) provide contrast-free or alternative pathways — defaulting to contrast in a contraindicated patient is the wrong, avoidable harm.

13. Proximal DVT → treat as VTE

WHAT YOU SEE

Step: proximal DVT on US → treat as VTE

WHAT IT ENCODES

DVT and PE are one disease (VTE). In a patient with suspected PE, finding a proximal DVT on compression US confirms VTE and justifies full anticoagulation without further chest imaging — useful when CTPA is contraindicated or non-diagnostic. Management is identical regardless of which territory is imaged. Mnemonic: 'Catch the clot in the leg, you've caught the PE — treat the same.'

BOARD TRAP

WRONG: 'Proximal DVT confirmed on US, but withhold anticoagulation until CTPA proves a PE.' Proximal DVT already establishes VTE — delaying treatment to chase imaging confirmation of PE risks fatal embolization; anticoagulate now.

14. Post-VTE sequelae (PTS / CTEPH)

WHAT YOU SEE

Ghost fleet labeled PTS, post-PE syndrome, CTEPH

WHAT IT ENCODES

Long-term follow-up watches for chronic complications: post-thrombotic syndrome (PTS — chronic leg pain, swelling, hyperpigmentation, venous ulcers from valvular damage after DVT) and chronic thromboembolic pulmonary hypertension (CTEPH — persistent exertional dyspnea months after PE). Screen suspected CTEPH with a V/Q scan (more sensitive than CTPA for chronic clot), then confirm with right heart catheterization; treat with pulmonary thromboendarterectomy. Mnemonic: 'After the clot: PTS in the leg, CTEPH in the lung.'

BOARD TRAP

WRONG: 'Work up suspected CTEPH (chronic dyspnea after PE) with CTPA as the first/best test.' V/Q scan is the preferred screening test for CTEPH (higher sensitivity for chronic/organized thrombus); a normal V/Q effectively excludes it, whereas CTPA can miss distal chronic disease.